PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Medical waste management-related factors affecting health and experiences of health risks among medical waste handlers in low- and middle-income countries: A systematic review protocol of qualitative studies
AUTHORS	Huda, Md. Nazmul; Hailemariam, Tewodros Getachew; Hossain, Syeda Zakia; Malo, James Sujit; Khan, Sajedul; Hadisuyatmana, Setho; Ferdous, Afsana; Akombi-Inyang, Blessing; Islam, Rakibul M; Renzaho, Andre

VERSION 1 – REVIEW

REVIEWER	Mistry, Sabuj University of New South Wales, Centre for Primary Health Care and Equity
REVIEW RETURNED	28-Sep-2021

GENERAL COMMENTS	The review article on " Experiences of health risks among medical waste handlers in low and middle-income countries: A systematic review protocol of qualitative studies" is well thought out, designed and structured which sheds light on the health risks among medical waste handlers. This review would be a timely piece in the current COVID-19 pandemic with increased medical products. I congratulate the authors' efforts for crafting this piece scientifically. Although the article is well-written, I suggest the authors address some minor issues listed below: Abstract 1. Page 4, line 7-9: Please change it to “Health risks associated with medical waste management (MWM) in low and middle-income countries (LMICs) are important public health concern.” 2. Page 4, line 16-20: “The review will answer the following questions: (i) what factors affect MWM in LMICs? and (ii) what are common health risks experienced by medical waste handlers in LMICs?” repetition, please remove. Background 1. Please define LMICS, and which you are going to include in the present study 2. Suggest revise this line, page 3, line 29-32: “The healthcare sector in low and middle-income countries (LMICs) is growing fast, due to rapid population growth and increased access to medical services, contributing to generating increased medical wastes” to “The healthcare sector is growing fast in low and middle-income countries (LMICs) due to rapid population growth and concomitant increased use of medical services are contributing to generating increased medical waste”

	3. Page 4, line 33-34: “Although studies of MWM and its health risks among medical waste handlers in LMICs are available”, can you cite some reference? 4. Page 4, line 34: “the studies are scattered in answering different research questions,”, can you mention what are those research questions you are refereeing to? Methods 1. Page 8, line 12: Please add a text box or table to present the inclusion and exclusion criteria 2. Is there any measure that you would like to take to minimize the potential risk of bias

REVIEWER	Velasco Garrido, Marcial Universitätsklinikum Hammburg-Eppendorf, Institute for Occupational and Maritime Health
REVIEW RETURNED	07-Oct-2021

GENERAL COMMENTS	The authors report a systematic review protocol for an interesting and relevant issue, probably affecting many workers in LMIC. The protocol is in general well written and reports all information needed to reproduce the planed review. I have two points to raise, which the authors should carefully consider before the protocol is published. My first concern is regarding the formulation of the main research question. The authors report their main question as "what factors affect MWM in LMICs?". For me this question is too general. What kind of factors do the authors have in mind, and first of all, what is meant with "affect"; affect health, quality of life, income, satisfaction with life...? The review protocol needs to specify this question more clearly. The second concern is regarding the search strategy proposed in the protocol. First, you say you will use subject headings (SH). I think your search would be more sensitivity whithout restrictions to SH, i.e. searching for free words/ext in Title and Abstract. Second, I am afraid, that including "Concept 4" (see page 17, line 3-13) with the boolean operator AND in the search strategy will restrict the sensitivity of the straegy too much. The first proposed research question - which needs more specification - is now formulated very general (factors affecting medical waste collectors). WIth step#4 of the proposed research you will restrict your search only to health issues, which is only the secondary research question. I recommend to run the search without this step. Otherwise, I am afraid you may loose a lot of relevant papers addressing the main research question (factors affecting MWM in LMIC). Minor issues; In page 6, line 25 I think the name of the mask should be KN95 instead of N95 Page 5, line 57, page 6 line 27, and abstract: it seems more appropiated to say "injuries and infections" instead "infections and injuries". Page 8, line 28-29, this sentence repeats another one before. Page 8, line 38, remove "??" Page 9, lines 14-18, "Interventions" seems to me not applicable to your review. Page 9, lines 47-49 I guess you mean the "study selection" instead the "search"

	When this issues are solved, the review protocol should be published. The authors report a systematic review protocol for an interesting and relevant issue, probably affecting many workers in LMIC. The protocol is in general well written and reports all information needed to reproduce the planed review. I have two points to raise, which the authors should carefully consider before the protocol is published. My first concern is regarding the formulation of the main research question. The authors report their main question as "what factors affect MWM in LMICs?". For me this question is too general. What kind of factors do the authors have in mind, and first of all, what is meant with "affect"; affect health, quality of life, income, satisfaction with life...? The review protocol needs to specify this question more clearly. The second concern is regarding the search strategy proposed in the protocol. First, you say you will use subject headings (SH). I think your search would be more sensitivity whithout restrictions to SH, i.e. searching for free words/ext in Title and Abstract. Second, I am afraid, that including "Concept 4" (see page 17, line 3-13) with the boolean operator AND in the search strategy will restrict the sensitivity of the straegy too much. The first proposed research question - which needs more specification - is now formulated very general (factors affecting medical waste collectors). WIth step#4 of the proposed research you will restrict your search only to health issues, which is only the secondary research question. I recommend to run the search without this step. Otherwise, I am afraid you may loose a lot of relevant papers addressing the main research question (factors affecting MWM in LMIC). Minor issues; In page 6, line 25 I think the name of the mask should be KN95 instead of N95 Page 5, line 57, page 6 line 27, and abstract: it seems more appropiated to say "injuries and infections" instead "infections and injuries". Page 8, line 28-29, this sentence repeats another one before. Page 8, line 38, remove "??" Page 9, lines 14-18, "Interventions" seems to me not applicable to your review. Page 9, lines 47-49 I guess you mean the "study selection" instead the "search" When this issues are solved, the review protocol should be published.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

Comments/Suggestions	**Authors' Responses**
Comments to the Author: The review article on " Experiences of health risks among medical waste handlers in low	We appreciate this comment.

Comments/Suggestions	Authors' Responses
and middle-income countries: A systematic review protocol of qualitative studies" is well thought out, designed and structured which sheds light on the health risks among medical waste handlers. This review would be a timely piece in the current COVID-19 pandemic with increased medical products. I congratulate the authors' efforts for crafting this piece scientifically. Although the article is well-written, I suggest the authors address some minor issues listed below:	
Abstract 1. Page 4, line 7-9: Please change it to "Health risks associated with medical waste management (MWM) in low and middle-income countries (LMICs) are important public health concern."	We have changes in the abstract (Page 2). Please see the changes below. Health risks associated with medical waste management (MWM) in low and middle-income countries (LMICs) are an important public health concern.
2. Page 4, line 16-20: "The review will answer the following questions: (i) what factors affect MWM in LMICs? and (ii) what are common health risks experienced by medical waste handlers in LMICs?" repetition, please remove.	We have deleted the suggested texts. Please see page 2 of the revised manuscript.
Background 1. Please define LMICS, and which you are going to include in the present study	We have taken the suggestions on board and have made revisions in the background. Below are the changes (Page 3). Low-income countries are those with a gross national income (GNI) per capita, calculated using the World Bank Atlas method, of $1,045 or less in 2020. Lower-middle and upper-middle-income countries are those with a GNI per capita from $1,046 to $4,095 and $4,096 to $12,695, respectively [4].

Comments/Suggestions	Authors' Responses
2. Suggest revise this line, page 3, line 29-32: "The healthcare sector in low and middle-income countries (LMICs) is growing fast, due to rapid population growth and increased access to medical services, contributing to generating increased medical wastes" to "The healthcare sector is growing fast in low and middle-income countries (LMICs) due to rapid population growth and concomitant increased use of medical services are contributing to generating increased medical waste"	We have taken the suggestion on board and have revised the sentence accordingly. Please see page 3.
3. Page 4, line 33-34: "Although studies of MWM and its health risks among medical waste handlers in LMICs are available", can you cite some reference?	We have provided references in the revised manuscript. Please see page 4.
4. Page 4, line 34: "the studies are scattered in answering different research questions,", can you mention what are those research questions you are refereeing to?	We appreciate this comment. We acknowledge that there were problems in wording of the sentence. Therefore, we have revised the sentence. Below is the modified sentence (Page 4). Although studies of MWM and its health risks among medical waste handlers in LMICs are available, the findings of these studies were not systematically synthesised.
Methods 1. Page 8, line 12: Please add a text box or table to present the inclusion and exclusion criteria	Thank you for the suggestion. We have provided a table to present the inclusion criteria of the proposed systematic review (pages 7-8).
2. Is there any measure that you would like to take to minimize the potential risk of bias	Yes, in line with existing literature, our review will use the Critical Appraisal Skills Programme (CASP) tool to assess methodological quality and the risk of bias. We have made the following revisions in the revised manuscript (Page 9). Studies included in this review will be assessed for methodological quality and risk of bias [22] using the

Comments/Suggestions	Authors' Responses
	Critical Appraisal Skills Programme (CASP) quality assessment tool for qualitative studies [23].

Reviewer 2

Comments/Suggestions	Authors' Responses
Comments to the Author: The authors report a systematic review protocol for an interesting and relevant issue, probably affecting many workers in LMIC. The protocol is in general well written and reports all information needed to reproduce the planed review.	We appreciate this comment.
I have two points to raise, which the authors should carefully consider before the protocol is published. My first concern is regarding the formulation of the main research question. The authors report their main question as "what factors affect MWM in LMICs?". For me this question is too general. What kind of factors do the authors have in mind, and first of all, what is meant with "affect"; affect health, quality of life, income, satisfaction with life...? The review protocol needs to specify this question more clearly.	Thank you for this suggestion. However, we respectfully disagree with the reviewer. After careful deliberation, we have decided to retain our research question. Our first review question was 'What are the factors influencing MWM in LMICs? We believe our research questions are reflective of our study objectives. Our first aim is to synthesis the key factors influencing MWM in LMICs. However, we have revised the first question (Page 2) to be more explanatory: 'What are the factors influencing MWM in LMICs? We have also made related changes throughout the manuscript (Pages 3-5, 7-10). As indicated above, in our review, we are not exploring factors of health outcomes e.g., quality of life, income, and satisfaction with life. We hypothesized that the factors that contribute to medical waste management in LMICs. The factors may include a lack of knowledge/awareness, coordination, skilled workers, training, resources etc., which will be investigated in our review. Secondly, we do not know yet what sorts of health problems/risks are experienced by the waste-handers

Comments/Suggestions	Authors' Responses
	in LMICs, which will be investigated in our second research question. Once a key health problem is addressed, the next step would be to identify factors associated with that particular health outcome/s, which is not a scope in our current review.
The second concern is regarding the search strategy proposed in the protocol. First, you say you will use subject headings (SH). I think your search would be more sensitivity without restrictions to SH, i.e. searching for free words/text in Title and Abstract. Second, I am afraid, that including "Concept 4" (see page 17, line 3-13) with the boolean operator AND in the search strategy will restrict the sensitivity of the strategy too much. The first proposed research question - which needs more specification - is now formulated very general (factors affecting medical waste collectors). WIth step#4 of the proposed research you will restrict your search only to health issues, which is only the secondary research question. I recommend to run the search without this step. Otherwise, I am afraid you may loose a lot of relevant papers addressing the main research question (factors affecting MWM in LMIC).	Thank you for this comment. Following this comment, we have revised the search strategy. Please see the additional file 1 – search strategy.
Minor issues; In page 6, line 25 I think the name of the mask should be KN95 instead of N95 Page 5, line 57, page 6 line 27, and abstract: it seems more appropriated to say "injuries and infections" instead "infections and injuries".	Thank you. We have corrected the suggested words (Page 4).
Page 8, line 28-29, this sentence repeats another one before.	We have deleted the sentence. Please see page 6.

Comments/Suggestions	Authors' Responses
Page 8, line 38, remove "??"	We have removed the "??". Please see page 7.
Page 9, lines 14-18, "Interventions" seems to me not applicable to your review.	We have deleted the suggested lines. Please see page 8.
Page 9, lines 47-49 I guess you mean the "study selection" instead the "search"	Thank you for this comment. We have corrected it. Please see page 9.
When this issues are solved, the review protocol should be published.	Many thanks for this compliment. We have addressed all issues suggested by you.

VERSION 2 – REVIEW

REVIEWER	Velasco Garrido, Marcial Universitätsklinikum Hammburg-Eppendorf, Institute for Occupational and Maritime Health
REVIEW RETURNED	17-Nov-2021

GENERAL COMMENTS	I raised two major concerns in my review, which I do not see fully addressed. Regarding the research question the authors have swapped the word "affect" with the word "influence". In my view the research question stills to general for a systematic review. The explanation ot the authors does not convince me. I still miss an specifiaction / concretization of the question. As it is now, the question "What are the factors inlfuencing MWM...?" I miss an explanation about "Influence MWM regarding to what?". The authors provide a list of potential factors... Other factors that may influence the MWM could be the wheather, the distance, the availability of roads.... I do not think the authors mean that. This is only to ilustrate what I meant with the need of a more preicse research question. The second point I raised was regarding the use of subject headings in the search strategy. The authors report they have revised the search strategy. I cannot see a singel change in the text regarding teh search strategy (pages 5 and 6). The still write about subject headings. Do the authors mean the are going to search the terms listed in the appendix in the field subject headings (SH)? If so, the search won´t be sensitive enough. Maybe the authors mean "search terms" instead of "subject headings"? but this is not clear from the text. The description of the search strategy should thus be more accurate. I still think the search terms should not be restricted to subjec headings but include title and abstract in order to achieve a good balance between sensitivity and specifity. instead of searching the terms as text in title or abstract. Again, the authors have not provided

VERSION 2 – AUTHOR RESPONSE

Reviewer 1: Mr. Marcial Velasco Garrido, Universitätsklinikum Hammburg-Eppendorf

Comments/Suggestions	Authors' Responses
Comments to the Author: I raised two major concerns in my review, which I do not see fully addressed. Regarding the research question the authors have swapped the word "affect" with the word "influence". In my view the research question still too general for a systematic review. The explanation of the authors does not convince me. I still miss an specification / concretization of the question. As it is now, the question "What are the factors influencing MWM...?" I miss an explanation about "Influence MWM regarding to what?". The authors provide a list of potential factors... Other factors that may influence the MWM could be the whether, the distance, the availability of roads.... I do not think the authors mean that. This is only to illustrate what I meant with the need of a more precise research question.	We thank the reviewer for taking the time to provide valuable comments. We have addressed the two concerns. Regarding the research question, as this review will be a comprehensive synthesis of qualitative studies on the experience of health risks among medical waste handlers, the objective is set to be general to include a broad spectrum of factors that can play a role in affecting medical waste handlers' health in LMIC settings. Following the reviewer's comment, we have revised the research question and provided context about the aspects that will be considered as factors affecting the health of medical waste handlers in LMICs. The factors include: 1. Individual-level factors such as training and knowledge; 2. System-level factors such as infrastructure and provision of technology/supplies to prevent health risk outcomes; and 3. Policy-level factors such as policy & regulations. Please see page 5 of the revised question. Changes have also been reflected throughout the manuscript. Please see pages 1-12.
The second point I raised was regarding the use of subject headings in the search strategy. The authors report they have revised the search strategy. I cannot see a single change in the text regarding the search strategy (pages 5 and 6). The still write about subject headings. Do the authors	As the reviewer mentioned, the searching strategies will not be restricted to the subject headings only. The search strategy will include all possible search terms, keywords and phrases relevant to the topic. As per a database of interest to conduct the search, we will apply a multi-

Comments/Suggestions	Authors' Responses
mean they are going to search the terms listed in the appendix in the field subject headings (SH)? If so, the search won´t be sensitive enough. Maybe the authors mean "search terms" instead of "subject headings"? but this is not clear from the text. The description of the search strategy should thus be more accurate. I still think the search terms should not be restricted to subject headings but include title and abstract in order to achieve a good balance between sensitivity and specify. Instead of searching the terms as text in title or abstract. Again, the authors have not provided this.	purpose (.mp) search across several fields, including title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept word, and unique identifier, to have efficient search outputs. Moreover, a consultation with a librarian will be carried out to apply efficient searching strategies. This has now been reflected in the manuscript (see the section 'search strategy', pages 5-6) and the additional file 2.

VERSION 3 – REVIEW

REVIEWER	Velasco Garrido, Marcial Universitätsklinikum Hammburg-Eppendorf, Institute for Occupational and Maritime Health
REVIEW RETURNED	28-Feb-2022

GENERAL COMMENTS	The authors have now fully addressed my points